

Pelvic architecture shapes load-dependent sacroiliac motion

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Abstract

Sex describes population differences in pelvic form, but its adequacy as a mechanical descriptor depends on how much information explicit anatomy adds. We examined sacroiliac joint responses in 278 imaging-derived pelvises using five standardized loading configurations and matched geometry–bone-density variants. Under three localized force pairs, age-adjusted female-minus-male translation coefficients ranged from +0.055 to +0.258 mm. Including pelvic volume, inlet depth, interspinous width and subpubic angle attenuated these coefficients to -0.031 to +0.013 mm, with all fully adjusted confidence intervals spanning zero. This attenuation does not establish sex equivalence or causal mediation. Preserving individual geometry while standardizing the bone-density field retained subject-specific translation, with RMSE no greater than 0.028 mm and minimum identity-line $R^2 = 0.946$ across loads. Load path nevertheless changed the response: redistributing the same total acetabular force from bilateral to unilateral support amplified asymmetry, and transverse force pairs applied at different sites produced distinct component patterns. Explicit pelvic architecture and loading thus provide a more informative description of modeled sacroiliac response than sex alone. The geometry–density comparison is conditional on the implemented density–modulus law, fixed cartilage and ligament properties and pretension, and passive static mechanics.

Keywords: sacroiliac joint; pelvic functional anatomy; pelvic biomechanics; sexual dimorphism; pelvic architecture; load transfer; finite-element analysis

1 Introduction

Small sacroiliac joint (SIJ) movements provide a way to examine how the pelvic ring transfers load. Articular geometry, ligamentous restraint and muscular compression contribute to stability, while the distribution of force determines which parts of the ring carry that load (Vleeming

et al., 2012; Snijders et al., 1993; Hammer et al., 2013). Describing joint motion therefore requires both an anatomical description and a defined mechanical context.

Human pelvic architecture varies in overall size and in the proportions of the inlet, mid-pelvis and pubic arch. Sex captures some of this variation, including developmental differences in obstetrically relevant dimensions (Huseynov et al., 2016), but combines several anatomical features into one category. A female–male difference in joint response could reflect pelvic size, inlet depth, interspinous width or subpubic geometry, as well as features not represented by those measurements. A group contrast alone cannot distinguish these possibilities.

The central anatomical question is whether sex retains an association with SIJ response once these dimensions are described explicitly. Comparing age-adjusted and anatomy-adjusted sex coefficients can establish how much the group contrast changes, without identifying a causal pathway. This distinction matters for functional interpretations of pelvic sexual dimorphism: the relationship between support and obstetric accommodation cannot be inferred from an unadjusted mobility difference (Pavličev et al., 2020).

A complementary question concerns the anatomical information needed to retain individual mechanical responses. Previous population modeling linked SIJ auricular morphology to its mechanical environment (Henyš and Hammer, 2025). Matched models that retain individual geometry while standardizing the bone-density field test whether that geometry also preserves joint-motion outputs. The reciprocal comparison, retaining individual density on a common geometry, provides a second reference. These comparisons address the represented bone-density variation, with soft-tissue properties held fixed.

Both questions require comparison of the same anatomies across load paths. Bilateral and unilateral acetabular loading redistribute a fixed total force, while localized force pairs probe different sites on the ring. The latter are motivated by obstetric anatomy but do not simulate childbirth, which additionally involves fetal contact, soft tissues and changing boundary conditions (Chen and Grimm, 2021). Here, using 278 imaging-derived pelves from the BoneDat workflow (Henyš and Kuchař, 2025), our primary aim was to determine how explicit pelvic dimensions alter sex-associated translation differences under localized loading. We then assessed how well individual geometry preserves responses under standardized bone density, using matched load comparisons to establish their mechanical context. Volume–motion scaling was a supporting, exploratory analysis. These questions and their directional hypotheses were retrospective and not preregistered.

2 Materials and methods

2.1 Cohort and provenance

The precomputed cohort comprised 278 pelvic models (150 female, 128 male), aged 16–91 years (mean $\pm$ SD: 53.9 ± 19.3 years), using the BoneDat morphology workflow (Henyš and Kuchař, 2025). Demographic, morphometric and size tables were joined by subject identifier. Eight morphometric variables and the kinematic outcomes were complete for the included cohort (Table 1). The previously recorded ethical approval is University Hospital Hradec Králové, reference 202411IO3P, with waiver of informed consent. Documentation of the cohort selection and its overlap with the earlier 281-model study (Henyš and Hammer, 2025) is required to distinguish reuse of imaging from independent validation.

Table 1: Demographic and morphometric characteristics. Values are mean $\pm$ SD; dimensions are measured from mapped landmarks.

Variable	Overall	Female	Male	Missing
N	278	150	128	0
Age (years)	53.90 ± 19.33	53.36 ± 19.74	54.52 ± 18.91	0
Inlet AP diameter (mm)	117.04 ± 10.52	121.04 ± 10.02	112.34 ± 9.07	0
Biacetabular width (mm)	125.81 ± 8.53	129.08 ± 8.21	121.98 ± 7.22	0
Biischiadic width (mm)	104.27 ± 9.33	109.87 ± 7.13	97.71 ± 7.06	0
Bituberous width (mm)	132.24 ± 9.15	136.46 ± 8.29	127.30 ± 7.49	0
Iliopectineal eminence width (mm)	102.38 ± 7.75	106.20 ± 7.02	97.90 ± 5.96	0
Pelvic inlet transverse (mm)	131.58 ± 8.88	134.85 ± 8.59	127.75 ± 7.61	0
Sacral width (mm)	117.08 ± 6.08	116.75 ± 5.67	117.47 ± 6.54	0
Subpubic angle (deg)	78.12 ± 10.12	85.80 ± 6.14	69.11 ± 5.26	0
Global scale factor (-)	0.96 ± 0.06	0.92 ± 0.05	1.00 ± 0.05	0
Total pelvic volume (cm ³)	823.58 ± 149.32	736.79 ± 105.99	925.28 ± 126.99	0
Total pelvic surface (cm ²)	1568.63 ± 169.26	1479.09 ± 133.64	1673.57 ± 144.80	0

2.2 Anatomical dimensions and their landmarks

Figure 1 shows all eight morphometric measures in Table 1, using the archived template landmarks from the measurement workflow. Each of the seven linear dimensions is the three-dimensional Euclidean distance $d = \|\mathbf{p}_2 - \mathbf{p}_1\|$, rather than its projection into a display plane. The subpubic angle uses the second of three landmarks as its vertex:

$$\alpha_{sub} = \arccos \left[\frac{(\mathbf{p}_1 - \mathbf{p}_2) \cdot (\mathbf{p}_3 - \mathbf{p}_2)}{\|\mathbf{p}_1 - \mathbf{p}_2\| \|\mathbf{p}_3 - \mathbf{p}_2\|} \right], \quad (1)$$

with numerical clipping to $[-1, 1]$ and conversion to degrees. The landmark labels are local to each measurement. Landmarks were mapped from the template to each subject; interpolation settings and archived numerical checks are given in Appendix A.1. Supplementary Figure S2 illustrates the three rotational coordinates, the definition of point translation, and the three bone bodies included in volume and surface-area summaries.

2.3 Finite-element model

The archived simulation metadata specify DOLFINx 0.10, 166,808 cells and 48,291 mesh nodes. Small-strain linear elasticity was solved with PETSc on a shared reference domain. Subject anatomy was represented by $\mathbf{x} = \mathbf{X} + \phi_i(\mathbf{X})$; the displacement map was interpolated using radial basis functions (RBFs). Spatial gradients and integration measures were pulled back to the reference domain (Bonet and Wood, 2008). This representation approximates individual anatomy on a shared discretization; it does not guarantee exact cortical boundaries or invertibility without Jacobian checks.

Experimental relationships between bone density and stiffness motivate density-dependent material assignment (Keller, 1994). The archived implementation used the following bone-modulus approximation:

$$E/\text{MPa} = \begin{cases} 2398, & \rho_{\text{ash}} \leq 0.486, \\ 10200\rho_{\text{ash}}^2, & \rho_{\text{ash}} > 0.486, \end{cases} \quad \rho_{\text{ash}} = (\rho + 0.09)/1.14, \quad (2)$$

where numerical density values are in g cm^{-3} and ρ denotes the mapped hydroxyapatite-density input. The equation, density conversion and low-density floor specify the implemented model and are not presented as a verbatim constitutive law from Keller. The constant low-density branch suppresses modulus variation below its threshold; the comparison must not be interpreted as an unrestricted test of bone quality. Bone Poisson ratio was 0.30. Both SIJ cartilage and pubic symphysis used $E = 1.0$ MPa and $\nu = 0.45$, as recorded in all three simulation archives. These nominal choices were not calibrated to individual subjects.

Ligaments were assembled as linear spring stiffness matrices with a geometric stiffness contribution from prescribed pretension and a pretension force vector. The implementation does not update a tension-only active set during the static solve. Bilateral anterior SIJ, posterior SIJ, interosseous, sacrospinous and sacrotuberous bundles and the pubic bundle were included.

Per-bundle stiffnesses were respectively 1400, 5600, 200, 3000, 3000 and 1000 N/mm; total pretension was 20 N per bundle. Each bundle’s parameters were divided among its segments. Bone–cartilage interfaces were conforming; frictional contact and muscle activation were not modeled.

2.4 Loads and constraints

The superior S1 facet was constrained using a surface penalty $\gamma = 10^6$ N/mm³ (Appendix A). Figure 2 shows the load application sites; Table 2 gives force vectors in the simulation’s global coordinates; these must be distinguished from the subsequently constructed pelvis-aligned reporting frame. SP2leg applies +400 N vertically to each acetabulum, and SP1leg applies +800 N to the right acetabulum. LAB1 applies opposing ML forces of ± 400 N to the internal ring surfaces; LAB2 applies the same signed pair to the ischial tuberosities; LAB3 applies opposing AP forces of ± 400 N at the pubis and sacrococcygeal junction. Inspection of the archived template places both left application patches at greater global x than their right counterparts; thus both ML pairs point outward in that template, consistent with outward loading at both sites. We name the cases by application site because a force direction does not establish the direction of joint motion.

Surface tractions were normalized by mapped surface area, giving identical prescribed force resultants across subjects. LAB force pairs have zero net force, whereas standing cases have an 800 N net vertical force balanced by the sacral constraint. Equal sums of force magnitudes do not imply equal moments, work, or physiological demand. The analyses compare responses to these specified load systems, not force-normalized compliance across equivalent activities. All load cases include the prescribed ligament pretension.

Table 2: Standardized loads in global simulation coordinates. LAB1–LAB3 are independent mechanical proxies.

Load case	Mechanical proxy	Applied total forces	Resultant
SP2leg	Bilateral standing	[0,0,+400] N on right AC notch and [0,0,+400] N on left AC notch	800 N cranio-caudal
SP1leg	Unilateral standing	[0,0,+800] N on right AC notch	800 N cranio-caudal
LAB1	Internal ring force pair	[+400,0,0] N on ring_contact_left and [-400,0,0] N on ring_contact_right	0 N net; bilateral ML pair
LAB2	Ischial force pair	[+400,0,0] N on left_ischium_tuber and [-400,0,0] N on right_ischium_tuber	0 N net; bilateral ML pair
LAB3	Outlet AP force pair	[0,-400,0] N on pubis_ins and [0,+400,0] N on SCJ	0 N net; AP pair

2.5 Reference configuration and SIJ kinematics

The kinematic analysis compares each unloaded mapped surface $\mathbf{x}_i^0 = \mathbf{X} + \phi_i$ with its loaded surface $\mathbf{x}_{i\ell}^1 = \mathbf{x}_i^0 + \mathbf{u}_{i\ell}$. Archived FE displacement fields were reprocessed without rerunning equilibrium solves; correspondence and pairing checks are documented in Appendix A.1.

Relative ilium-to-sacrum motion was estimated using trimmed Kabsch registration of opposing surface regions within 5 mm, with a minimum of 80 nodes per region. The underlying least-squares rigid fit follows Kabsch (1976); trimming and ROI selection are implementation choices. Before registration, the inverse least-squares affine sacral map was applied to all loaded points. This removes sacral affine strain as well as rigid drift; the resulting quantities are affine-corrected relative motion, rather than unmodified physical joint displacement. The available rigid-only sensitivity check is reported below.

Motion was expressed in a pelvis-aligned ML/AP/CC frame, not a local articular normal/tangent frame. Fixed-axis Cardan xyz angles satisfy $R = R_z R_y R_x$. The bilateral mean of the Euclidean norms of these angle triplets is denoted $|\boldsymbol{\theta}|$; this is a small-angle summary, not the exact invariant rotation angle. Translation $|\mathbf{d}|$ is the bilateral mean of the norms of relative displacements evaluated at the sacral ROI centroid. Component summaries are bilateral means of absolute values. In particular, $|\theta_x|$ does not distinguish nutation from counternutation, and $|d_{ML}|$ is not a measurement of joint gapping. Translation-magnitude asymmetry is $|\|\mathbf{d}_L\| - \|\mathbf{d}_R\||$.

2.6 Matched variants and variance ratios

The full variant uses subject-specific geometry and bone density. The shape-only variant uses subject geometry and the spatially varying pointwise population-median density field; it is not homogeneous bone. The material-only variant uses subject density on the template geometry. Cartilage and ligament parameters remain nominal in all variants.

For each load and endpoint we calculated $100V_{shape}/V_{full}$ and $100V_{material}/V_{full}$, with 2000 paired subject bootstrap samples (Efron and Tibshirani, 1994). These non-additive ratios can exceed 100% and neither estimate causal variance fractions nor separate geometry-material interactions. Shape/full agreement was also assessed by RMSE, MAE and identity-line $R^2 = 1 - \sum_i (Y_{shape,i} - Y_{full,i})^2 / \sum_i (Y_{full,i} - \bar{Y}_{full})^2$.

2.7 Statistical analysis

The hypotheses organize a retrospective analysis and were not preregistered. The primary anatomical question (H3) was whether including size and pelvic dimensions attenuates age-adjusted female translation differences under localized loading. The supporting geometry comparison (H4) asked whether individual geometry retains between-subject variation and individual responses under standardized bone density. Two contextual predictions were that unilateral support increases motion and bilateral asymmetry (H1), and that both transverse force pairs increase the absolute ML component while reducing the CC component relative to bilateral support (H2); the AP pair was assessed separately. Volume–motion associations were exploratory. Distributions are summarized by medians and interquartile ranges. Paired component contrasts use $\Delta_{i\ell,k} = |d_k|_{i\ell} - |d_k|_{i,SP2leg}$, with 3000 bootstrap samples for median confidence intervals and two-sided exact binomial sign tests of the median contrast (zero differences excluded). Wilcoxon signed-rank statistics (Wilcoxon, 1945) are retained in the machine-readable output as a sensitivity diagnostic, but are not used for the primary inference because asymmetric differences can yield a signed-rank result that disagrees with the direction of the median. Benjamini–Hochberg correction was applied to the 12 component comparisons (Benjamini and Hochberg, 1995). H1 additionally uses paired tests for rotation magnitude, translation magnitude and translation asymmetry (a separate three-test family).

For each LAB case, nested OLS models used HC3 covariance (MacKinnon and White, 1985):

$$M0 : Y = \beta_0 + \beta_s \text{Female} + \beta_a \text{Age}_z + \epsilon, \quad (3)$$

$$M1 : Y = M0 + \beta_v \log(V)_z, \quad (4)$$

$$M2 : Y = M1 + \beta_{AP} \text{AP}_z + \beta_b \text{Biischadic}_z + \beta_p \text{Subpubic}_z. \quad (5)$$

Here model addition denotes addition to the linear predictor. Female is coded 1 and male 0; continuous predictors are standardized using cohort sample SDs. M2 contains no sex–volume interaction. FDR correction covers its six sex coefficients (two outcomes by three loads). M0/M1 and coefficient attenuation are descriptive. VIFs quantify collinearity without treating a threshold as proof of identifiability. The triad describes inlet depth, interspinous midpelvic width and the anterior pubic arch; it was selected for this retrospective analysis. Pooled and sex-stratified Spearman correlations are exploratory, not evidence of mediation.

Pooled OLS models include load, sex, age, log volume, the morphometric triad, load–

181 volume interactions and a sex–volume interaction, with covariance clustered by the 278 subjects
182 (Cameron and Miller, 2015). A main volume coefficient therefore refers to male SP2leg, not an
183 overall pooled slope. FDR is applied within each outcome’s pooled coefficients. Robust Wald
184 statistics are not converted to partial variance-explained measures.

185 Exploratory load-specific models regress $\log Y$ on $\log V$, sex, their interaction, age and the
186 triad. Male and female slopes are β_v and $\beta_v + \beta_{v:s}$; HC3 covariance supplies 95% Wald intervals.
187 FDR covers 20 sex-specific slope tests, with a separate family of ten interaction tests. Failure
188 to reject an interaction is not evidence of equal slopes. The alternative $\log(\text{scale})$ model is an
189 algebraic reparameterization because $\log V = 3 \log(\text{scale}) + \log V_{\text{template}}$; it is not an independent
190 robustness test. Sample confidence intervals do not include uncertainty in loads, constitutive
191 choices, segmentation or boundary conditions.

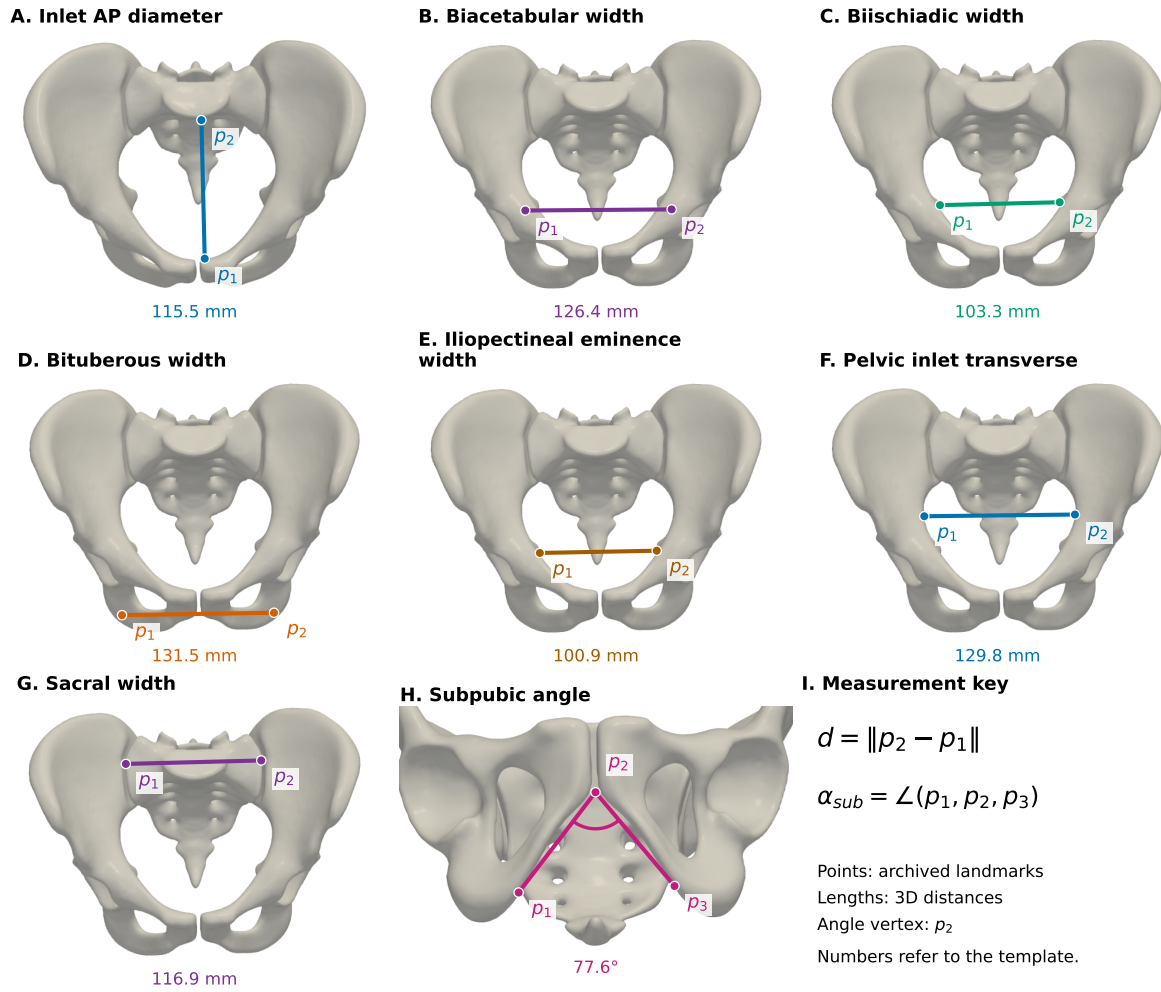


Figure 1: **Anatomical landmarks for all morphometric measures.** (A–G) Archived landmark pairs define the seven linear dimensions in Table 1. (H) The subpubic angle has vertex p_2 and arms toward p_1 and p_3 ; the view is normal to their plane. (I) Measurement definitions and landmark key. Markers and segments are overlaid on the reference pelvic geometry to keep internal landmarks visible. Values describe the template, not cohort averages. Lengths are measured in three dimensions; each camera direction is perpendicular to its measurement segment. Panels use different views and magnifications.

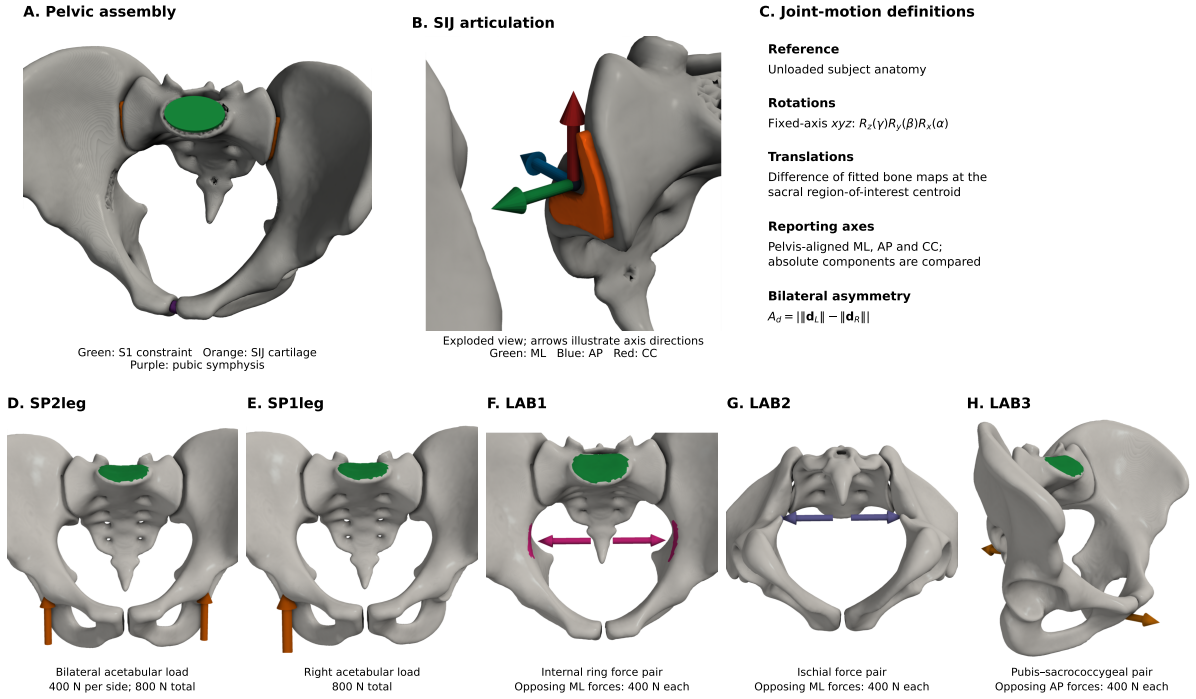


Figure 2: **Anatomy, motion definitions and load application.** (A) Pelvic assembly: green denotes the S1 constraint, orange SIJ cartilage and purple the symphysis. (B) Exploded articulation with illustrative axis directions; these arrows do not define the subject-specific PCA frame or anatomical signs. (C) Rotational and translational reporting conventions. (D,E) Bilateral and unilateral acetabular loads, each totaling 800 N. (F–H) Independent internal-ring, ischial and AP force pairs. Load arrows show applied global directions, not measured joint opening. The S1 penalty approximates a fixed boundary.

Each of the 278 anatomies contributed responses to all five loads (1390 subject–load observations) in each of the three geometry–density variants. This matched design first establishes the loading context for the anatomical comparisons.

3.1 Unilateral support amplifies asymmetry at unchanged total force

Redistributing a fixed total acetabular force changed the SIJ response. Under SP2leg and SP1leg, median rotation norms were 0.956° and 2.040° , and median translations were 1.058 and 1.627 mm, respectively. Median bilateral translation asymmetry was 0.065 mm in SP2leg and 1.602 mm in SP1leg (Fig. 3). The paired SP1leg-minus-SP2leg median difference in rotation norm was $+1.048$ (95% CI $+1.004$ to $+1.102$) degrees, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation magnitude was $+0.582$ (95% CI $+0.559$ to $+0.603$) mm, $q < 0.001$. The paired SP1leg-minus-SP2leg median difference in translation asymmetry was $+1.509$ (95% CI $+1.445$ to $+1.568$) mm, $q < 0.001$. Redistributing the same 800 N total force

therefore increased motion and, most prominently, bilateral asymmetry.

Table 3: SIJ motion summaries across 278 subjects. Values are median [interquartile range]; component columns are absolute magnitudes.

Load case	N	Rotation magnitude (deg)	Translation magnitude (mm)	AP translation component (mm)	ML translation component (mm)	CC translation component (mm)
SP2leg	278	0.956 [0.813, 1.109]	1.058 [0.871, 1.207]	0.366 [0.241, 0.482]	0.083 [0.054, 0.123]	0.952 [0.818, 1.112]
SP1leg	278	2.040 [1.765, 2.289]	1.627 [1.430, 1.869]	0.440 [0.357, 0.520]	0.235 [0.164, 0.345]	1.428 [1.213, 1.644]
LAB1	278	0.221 [0.188, 0.268]	0.154 [0.121, 0.190]	0.060 [0.044, 0.079]	0.077 [0.063, 0.092]	0.105 [0.073, 0.144]
LAB2	278	0.724 [0.649, 0.812]	0.255 [0.204, 0.328]	0.176 [0.099, 0.262]	0.134 [0.115, 0.154]	0.075 [0.049, 0.118]
LAB3	278	0.844 [0.672, 1.057]	0.574 [0.448, 0.707]	0.372 [0.260, 0.515]	0.092 [0.069, 0.115]	0.387 [0.326, 0.477]

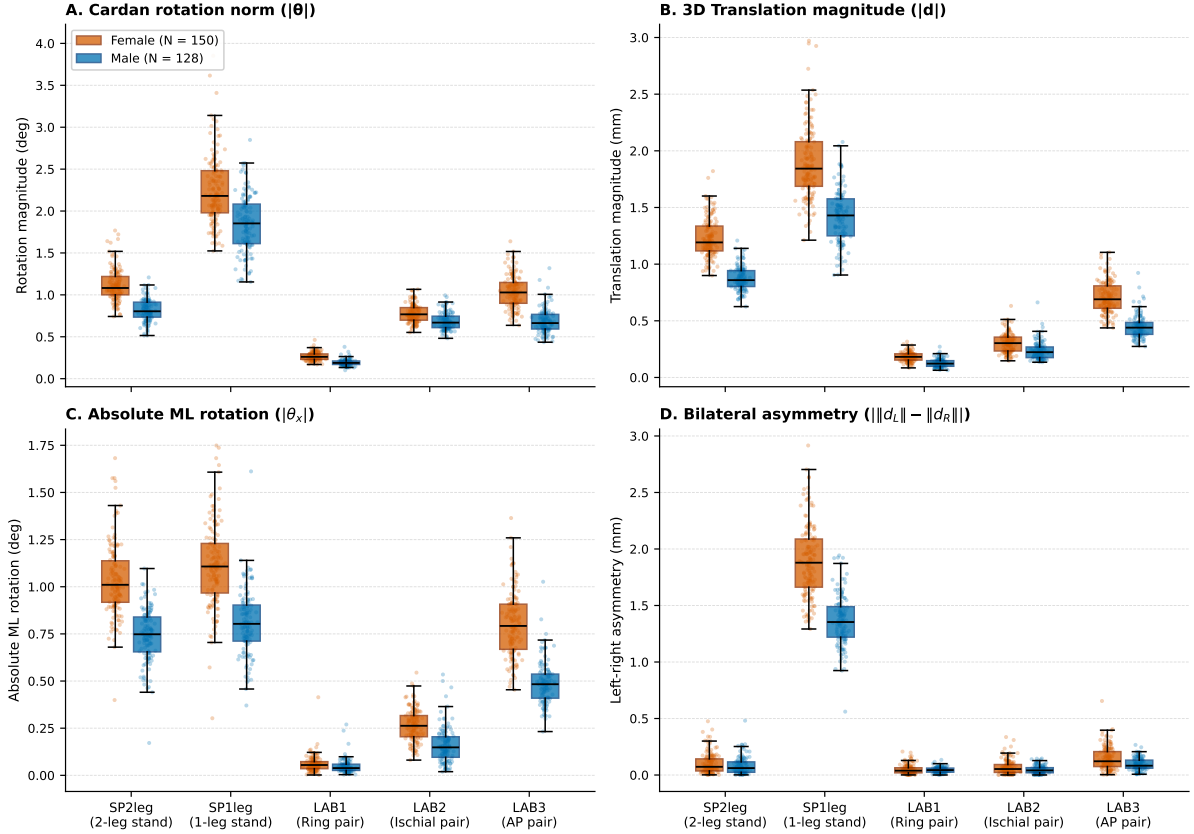


Figure 3: **Load-dependent motion and asymmetry.** Rotation-triplet norm, relative translation norm, absolute ML rotation and bilateral translation-magnitude asymmetry. Points are subjects; boxplots show medians and IQRs, with whiskers at 1.5 IQR. Female and male groups use vermilion and blue. Jitter is deterministic. No signed nutation direction is inferred.

3.2 Load application site selects distinct motion components

The localized force pairs separated the response into distinct component patterns (Fig. 4). Both transverse pairs reduced the absolute CC component relative to bilateral support, but the ML component increased only under LAB2: its paired median contrast was +0.052 mm (95% CI +0.047 to +0.061, $q < 0.001$), compared with −0.007 mm for LAB1 (95% CI −0.018 to +0.002, $q = 0.254$). Both also reduced the AP component. LAB3 increased the ML and AP

components while reducing the CC component. The shared transverse-redistribution prediction was therefore only partly supported. Supplementary Table S1 gives all paired contrasts and adjusted probabilities. Component magnitudes quantify motion along the reporting axes; the arrows connect componentwise median contrasts and are not physical trajectories.

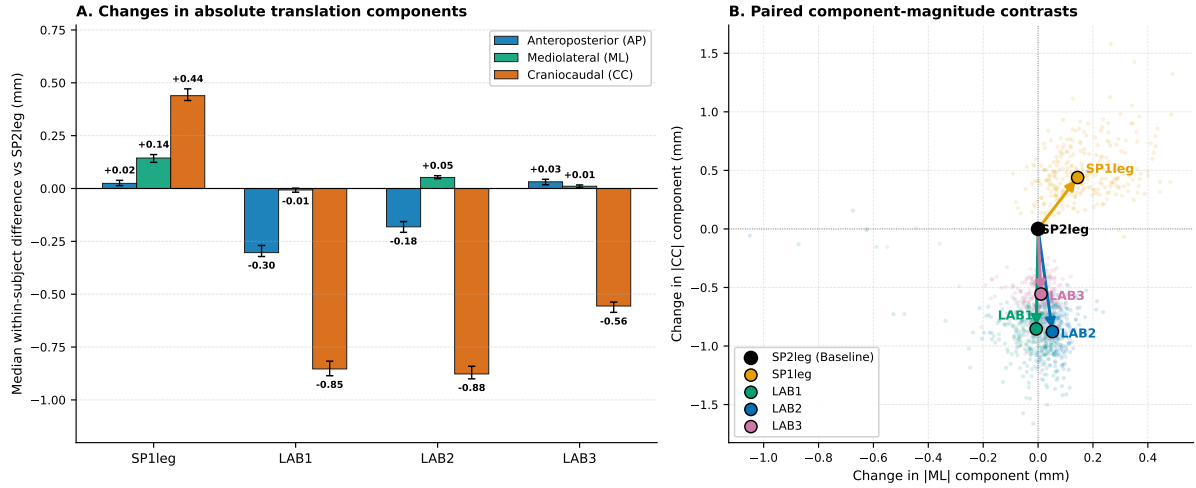


Figure 4: **Paired absolute-component contrasts.** (A) Median within-subject differences from SP2leg, with bootstrap 95% CIs. (B) Individual ML/CC contrast pairs and arrows to their componentwise medians. Positive values mean larger component magnitude, not outward joint motion. Corrected q values are reported in Supplementary Table S1; intervals are pointwise, not simultaneous.

3.3 Pelvic dimensions attenuate sex-associated translation differences

Explicit dimensions substantially attenuated the age-adjusted sex contrast under all three localized loads (Fig. 5). LAB1 female-minus-male translation coefficients were +0.055, +0.040 and +0.013 mm in M0, M1 and M2. The M2 interval was -0.005 to +0.031 mm ($p = 0.166$, $q = 0.566$). LAB2 female-minus-male translation coefficients were +0.071, +0.041 and -0.031 mm in M0, M1 and M2. The M2 interval was -0.078 to +0.016 mm ($p = 0.192$, $q = 0.566$). LAB3 female-minus-male translation coefficients were +0.258, +0.212 and +0.002 mm in M0, M1 and M2. The M2 interval was -0.060 to +0.064 mm ($p = 0.950$, $q = 0.950$). All three fully adjusted translation intervals spanned zero. The maximum non-intercept VIF in M2 was 5.55. Fully adjusted rotation intervals also spanned zero under all three loads (Supplementary Table S3). The exploratory subpubic-angle comparison distinguished the pooled association from those within sex groups. For LAB1, the pooled Spearman correlation between subpubic angle and translation magnitude was $r_s = 0.444$ ($p < 0.001$). The male-only correlation was $r_s = -0.063$ ($p = 0.480$). The female-only correlation was $r_s = -0.145$ ($p = 0.077$). These exploratory cor-

relations are uncorrected for multiplicity and do not estimate a geometric mechanism. Figure 5 and Supplementary Table S3 report M2 estimates for both outcomes.

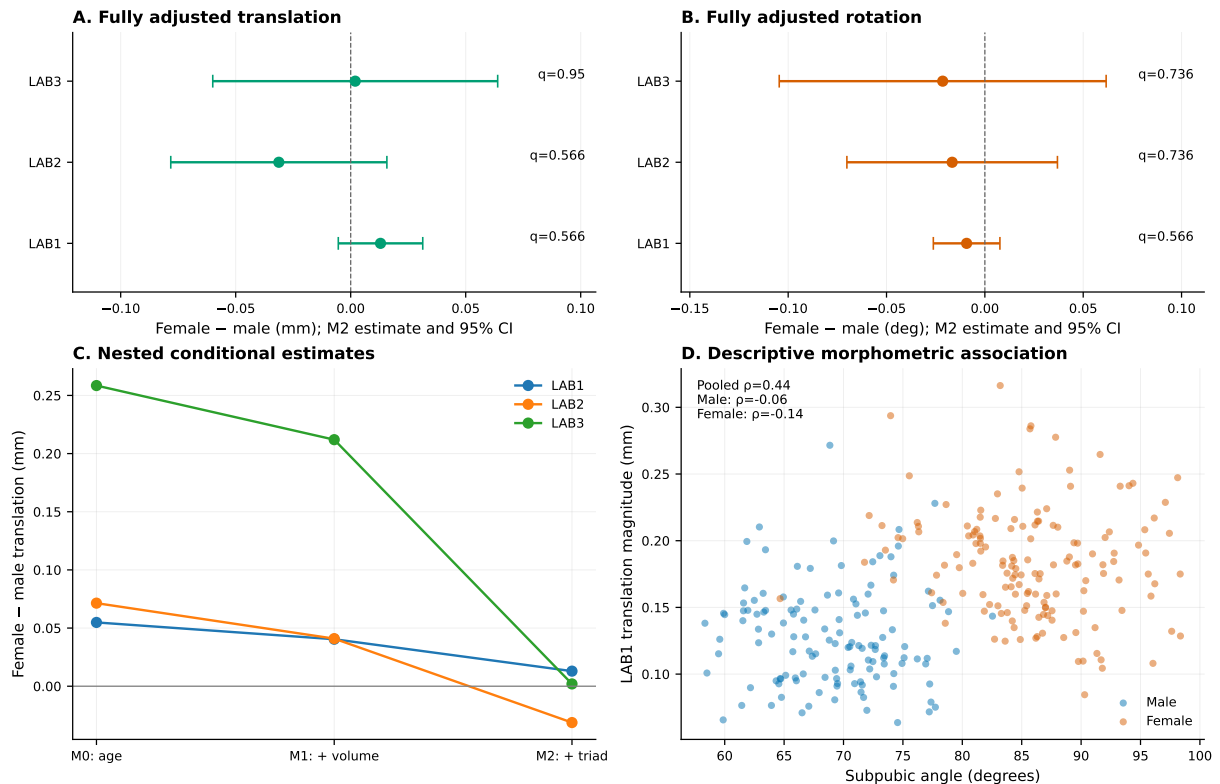


Figure 5: **Explicit anatomy attenuates sex-associated translation differences.** (A,B) M2 female-minus-male coefficients with HC3 95% CIs, shown on separate translation and rotation axes. Labels give BH-adjusted q values across six tests. (C) Translation coefficients attenuate as volume and dimensions enter the model; connecting lines compare specifications and are not causal pathways. (D) The positive pooled LAB1 subpubic-angle association is not reproduced within either sex. All M2 intervals span zero; this does not establish equivalence. M0 adjusts for age, M1 adds volume, and M2 adds the morphometric triad.

3.4 Individual geometry preserves responses under standardized bone density

Preserving individual geometry while standardizing the bone-density field retained individual motion closely. Across loads, translation had a minimum shape/full identity-line R^2 of 0.9457 and maximum RMSE of 0.0280 mm. For rotation, the corresponding values were 0.8777 and 0.0318 degrees. Across five loads and five endpoints, shape-only/full variance ratios ranged from 96.43% to 120.61% (median 100.68%), compared with 0.014% to 5.789% (median 0.244%) for material-only/full ratios. These are model-variant comparisons, not an additive variance decomposition (Fig. 6; Supplementary Table S2).

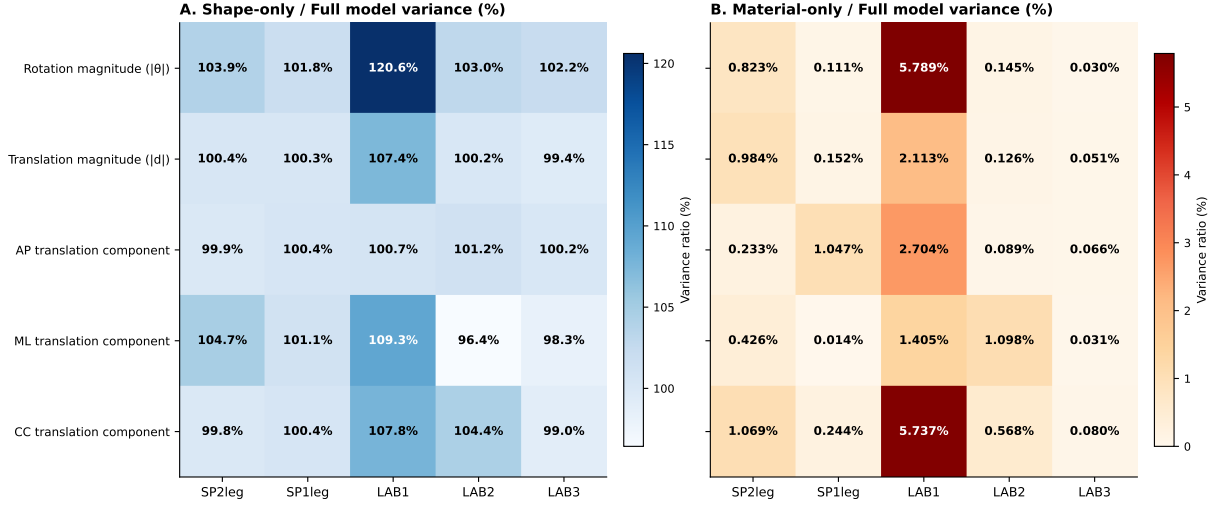


Figure 6: **Retaining individual geometry preserves between-subject variation.** (A) Shape-only/full and (B) material-only/full sample-variance ratios, in percent. Five endpoints are shown. Color scales differ between panels and follow their actual data ranges. Values above 100% are allowed. Subject-level agreement complements these variance summaries (translation RMSE ≤ 0.0280 mm; minimum identity-line $R^2 = 0.9457$). Bootstrap intervals and agreement diagnostics are provided in Supplementary Table S2; neither panel represents variance explained by an independent causal factor.

3.5 Larger pelves show smaller motions across loading configurations

In the supporting allometric analysis, larger pelvic volume was associated with smaller motion: all 20 sex-specific slopes were negative and survived FDR correction. The exponents ranged from -1.062 to -0.366; the largest adjusted slope q was 0.002. The inverse association with pelvic volume was therefore consistent across all five loading configurations. Full estimates and intervals are in Table 4. Sex-by-volume interaction q values ranged from 0.343 to 0.699; nonsignificance does not establish equivalence. Figure 7 and Table 4 use the same adjusted models. Pooled cluster-robust coefficients are reported in Supplementary Table S5.

Table 4: Adjusted log-log exponents by load and sex. Slope q values use the 20-test family; interaction q values use a separate ten-test family.

Load	Outcome	Male exponent [95% CI]	Male q	Female exponent [95% CI]	Female q	Interaction q
SP2leg	Rotation (deg)	-0.679 [-0.890, -0.469]	< 0.001	-0.605 [-0.764, -0.446]	< 0.001	0.699
SP2leg	Translation (mm)	-0.366 [-0.531, -0.200]	< 0.001	-0.402 [-0.522, -0.282]	< 0.001	0.699
SP1leg	Rotation (deg)	-0.622 [-0.895, -0.349]	< 0.001	-0.550 [-0.760, -0.339]	< 0.001	0.699
SP1leg	Translation (mm)	-0.460 [-0.711, -0.208]	< 0.001	-0.397 [-0.591, -0.203]	< 0.001	0.699
LAB1	Rotation (deg)	-0.916 [-1.144, -0.687]	< 0.001	-1.062 [-1.233, -0.891]	< 0.001	0.588
LAB1	Translation (mm)	-0.568 [-0.925, -0.210]	0.002	-1.003 [-1.270, -0.737]	< 0.001	0.343
LAB2	Rotation (deg)	-0.541 [-0.714, -0.367]	< 0.001	-0.674 [-0.804, -0.544]	< 0.001	0.494
LAB2	Translation (mm)	-0.818 [-1.179, -0.456]	< 0.001	-0.947 [-1.287, -0.606]	< 0.001	0.699
LAB3	Rotation (deg)	-0.850 [-1.048, -0.651]	< 0.001	-1.045 [-1.240, -0.851]	< 0.001	0.494
LAB3	Translation (mm)	-0.700 [-0.922, -0.478]	< 0.001	-0.904 [-1.159, -0.649]	< 0.001	0.494

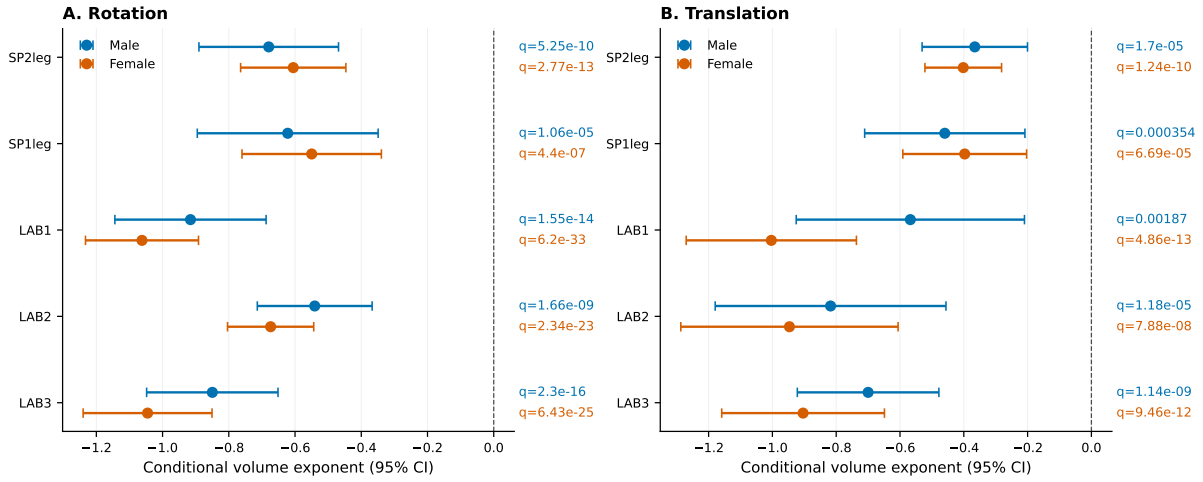


Figure 7: **Conditional log-volume slopes.** Male and female exponents with HC3 95% CIs from the same adjusted log-log models as Table 4. The vertical zero line indicates no conditional association. Confidence intervals are pointwise; labels give slope q values from the 20-test family. Raw scatterplots and adjusted conditional geometric-mean curves are supplied as Supplementary Figure S1.

3.6 Sensitivity of motion extraction

In six subjects selected at approximately evenly spaced archive indices (30 subject-load comparisons), replacing affine sacral correction with rigid-only correction changed bilateral rotation norms by at most 0.0229° and translation magnitudes by at most 0.0055 mm. This subset diagnostic does not bound errors in the remaining cohort (Supplementary Table S6).

4 Discussion

Explicit pelvic architecture provides a more informative description of modeled SIJ response than sex alone, and that response depends on how the ring is loaded. Under localized loading, including pelvic size and dimensions substantially attenuated the age-adjusted female translation coefficients. A separate comparison showed that preserving individual geometry under a standardized bone-density field retained much of the subject-specific response. Together, these results connect the interpretation of sex contrasts to the anatomical information required by the model.

Why sex is an incomplete mechanical descriptor

The female and male groups differed in several dimensions: female pelvises had lower total bone volume but greater mean inlet depth, interspinous width and subpubic angle (Table 1). Sex

therefore summarizes a combination of size and proportions rather than one mechanical feature. Adding those dimensions reduced the female translation coefficients from +0.055 to +0.258 mm across localized loads to -0.031 to +0.013 mm. All fully adjusted translation intervals included zero. This supports interpreting the initial group contrasts in relation to explicit anatomy; it neither establishes mechanical equivalence between sexes nor demonstrates causal mediation by the selected dimensions.

The subpubic-angle comparison exposes a related difficulty with aggregate anatomical associations. The positive pooled correlation with LAB1 translation was not reproduced within either sex: both within-sex estimates were weakly negative and their tests did not reject zero correlation. Thus the pooled pattern cannot be treated as evidence that increasing the angle predicts increasing motion within women or men. These exploratory results favor describing dimensions jointly and examining group structure. The selected triad is not a complete representation of pelvic architecture, and coefficient attenuation alone does not demonstrate improved prediction in new individuals. External assessment should examine anatomical overlap between sexes, collinearity and performance beyond this cohort.

Geometry retains the subject-specific mechanical response

The matched geometry–density comparison asks a different question from the sex models: how much of an individual’s modeled response survives a controlled simplification? Retaining individual geometry with a spatially varying population-median density field gave translation RMSE no greater than 0.028 mm and a minimum identity-line R^2 of 0.946. Rotation agreement was weaker but still close, with minimum $R^2 = 0.878$. The near-100% shape/full variance ratios and small material-only/full ratios were consistent with these paired comparisons. Individual geometry may therefore be a useful priority when constructing models for these particular joint-motion endpoints; acceptable density simplification depends on the intended error tolerance.

This hierarchy is conditional on the passive static model and its implemented density–modulus law, including a low-density modulus floor. Cartilage properties, ligament properties and pretension were fixed. Ligament pretension can substantially affect SIJ predictions (Heyland et al., 2025), and introducing individual soft-tissue variation could change the relative contributions. The variance ratios compare distinct mechanical systems and do not partition causally explained variance. The practical result is a tested simplification of the represented bone-density field, not evidence that material properties generally have little mechanical relevance.

Pelvic mobility depends on the load path

Anatomy did not determine a single, context-independent mobility value. Redistributing 800 N from bilateral to unilateral acetabular support increased translation and rotation and strongly amplified bilateral asymmetry. The total applied force was unchanged, but its moments and route through the ring changed. This gives a population-level mechanical context to established accounts of SIJ load transfer (Snijders et al., 1993; Vleeming et al., 2012).

Application site was also consequential. Both transverse force pairs reduced the craniocaudal component relative to bilateral support, yet only the ischial pair increased the mediolateral component after multiplicity correction. The complete shared transverse-redistribution hypothesis was not supported. Its failure shows why a common global loading direction is insufficient to specify the joint response. Anatomical dimensions should therefore be interpreted together with the location and distribution of force, rather than used to assign an intrinsic scalar mobility to a pelvis.

Size provides a common scaling context

The consistently negative conditional volume slopes provide supporting evidence that size influences response magnitude across load paths. Their direction is partly compatible with dimensional expectations: geometrically similar linear-elastic solids under fixed force and modulus give $u \propto L^{-1}$ and $\theta \sim u/L \propto L^{-2}$, corresponding to volume exponents of $-1/3$ and $-2/3$. Fixed spring parameters, heterogeneous tissues and varying proportions prevent these benchmarks from being exact predictions here. The fitted slopes also condition on other pelvic dimensions; they describe residual size associations under prescribed loading, not a universal biological scaling law. Undetected sex-by-volume interactions do not demonstrate equal slopes.

Implications for functional anatomy

Categorical sex labels describe population anatomy, but explicit dimensions connect that description more directly to a specified mechanical response. Beyond previous work on SIJ auricular morphology (Henyš and Hammer, 2025), the present comparisons link whole-pelvis dimensions to sex-associated motion differences and test whether individual responses persist when bone density is standardized. Their functional implication is that pelvic form should be evaluated through its interaction with load path: the same architecture can express different component patterns under different demands.

This interpretation offers a structural question for obstetric anatomy without treating the localized probes as childbirth simulations. Neither absolute mediolateral motion nor the subpubic-angle association measures signed joint opening or birth-canal expansion. Establishing those quantities would require direct surface-separation and canal measurements, alongside the fetal and soft-tissue interactions represented in childbirth models (Chen and Grimm, 2021). No delivery outcome or evolutionary trade-off follows from the present contrasts.

Limitations and experimental tests

The model represents passive, static mechanics with a constrained sacrum, nominal cartilage properties and linearized ligaments, without muscle activation or frictional contact. The density comparison inherits these choices. The cohort spans ages 16–91 and does not specifically represent pregnant or laboring women; pregnancy-related adaptation, parity, pain and delivery outcomes were not analyzed. The retrospective selection of dimensions and comparisons also limits confirmatory interpretation.

Affine sacral correction removes sacral strain as well as rigid drift. The six-subject rigid-only sensitivity check gave small changes in the reported endpoints, but cannot establish a cohort-wide error bound or validate their physiological magnitude. Sampling intervals exclude constitutive, mapping and boundary-condition uncertainty. Direct validation against cortical strains has been demonstrated for a subject-specific pelvic model (Anderson et al., 2005); validation of that endpoint cannot be transferred to SIJ kinematics here. Mesh and mapping-quality assessment and independent measurements under matched loads remain necessary, particularly given the difficulty of measuring small SIJ motions (Goode et al., 2008).

The resulting experimental predictions are specific: unilateral loading should increase bilateral asymmetry at matched total acetabular force, and transverse forces at the internal ring and ischial tuberosities should produce different component responses. Tests should measure signed SIJ surface separation alongside the reported point-motion endpoints. Comparing subject-specific and standardized density assignments against those measurements would test the proposed simplification, while introducing measured cartilage, ligament and pretension variation would test its scope. Independent specimens and loading configurations are needed to determine how far the anatomical relationships generalize.

5 Conclusion

Within these passive static comparisons, sex alone was an incomplete descriptor of sacroiliac response. Explicit pelvic size and dimensions substantially attenuated sex-associated translation differences, while individual geometry retained much of the modeled response under a standardized bone-density field and fixed soft-tissue properties. The resulting motion depended on load distribution and application site. Pelvic function is therefore more informatively described through individual architecture interacting with loading than through a fixed sex-specific mobility value; this interpretation does not establish sex equivalence, causal mediation or physiological validation.

A Mathematical formulation and verification

A.1 Archived implementation and correspondence checks

Subject landmarks were obtained by thin-plate-spline RBF interpolation of template-to-subject displacements (10 neighbors; smoothing parameter 100). These morphometric interpolation settings differ from those used for the FE surface reconstruction. Recomputing the eight measures for all 278 subjects reproduced the archived values within 10^{-8} mm or degrees.

Archived displacement fields were reprocessed without new equilibrium solves. Reference-to-reference checks were below 10^{-8} in degrees and mm for every subject. Node ordering was checked against all five archived reference displacement fields. Geometry and displacement were evaluated with the same RBF parameters used by the original workflow. Checkpoints preserve subject row indices consistently across the three variants. Pairing follows the common indexed shape/density inputs and the archived processing order; the auxiliary archives do not contain an independent subject-ID manifest.

A.2 Reference mapping and equilibrium

Let Ω_0 be the FE template and $\chi_i(\mathbf{X}) = \mathbf{X} + \phi_i(\mathbf{X})$ the unloaded subject geometry. The anatomical mapping and the elastic displacement are different fields. With $\mathbf{F}_i = \mathbf{I} + \nabla_X \phi_i$ and $J_i = \det \mathbf{F}_i$, the physical small-strain tensor is

$$\varepsilon_i(\mathbf{u}) = \text{sym}(\nabla_X \mathbf{u} \mathbf{F}_i^{-1}). \quad (6)$$

382 The pullback requires an orientation-preserving, nonsingular map. RBF smoothness alone does
 383 not prove this condition. Here $\nabla_X \mathbf{u}$ differentiates the displacement represented on the template;
 384 $\text{sym } \mathbf{B} = (\mathbf{B} + \mathbf{B}^T)/2$. Volume integration satisfies $dV_i = J_i dV_0$. Thus, for example, the elastic
 385 term on the template is $\int_{\Omega_0} (\mathbf{C}_i \boldsymbol{\varepsilon}_i(\mathbf{u})) : \boldsymbol{\varepsilon}_i(\mathbf{v}) J_i dV_0$. For physical coordinates in mm, find the
 386 discrete displacement $\mathbf{u}_h$ such that, for every test displacement $\mathbf{v}_h$, the weak problem is

$$387 \quad \int_{\Omega_i} \mathbf{C}_i \boldsymbol{\varepsilon}(\mathbf{u}) : \boldsymbol{\varepsilon}(\mathbf{v}) dV + \int_{\Gamma_{S1,i}} \gamma \mathbf{u} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{K}_{lig,i} \mathbf{u}_h = \int_{\Gamma_{load,i}} \mathbf{t} \cdot \mathbf{v} dS + \mathbf{v}_h^T \mathbf{f}_{pre,i}. \quad (7)$$

388 Here $\mathbf{C}_i$ is the isotropic elasticity tensor, $\mathbf{t}$ is the prescribed surface traction, and $\mathbf{u}_h, \mathbf{v}_h$ are nodal
 389 coefficient vectors in the spring terms. In each tissue, $\boldsymbol{\sigma} = 2\mu\boldsymbol{\varepsilon} + \lambda \text{tr}(\boldsymbol{\varepsilon})\mathbf{I}$, with $\mu = E/[2(1+\nu)]$
 390 and $\lambda = E\nu/[(1+\nu)(1-2\nu)]$. The surface penalty has units N/mm³, because its associated
 391 traction is $\gamma\mathbf{u}$ in N/mm². It imposes an approximate fixed boundary, not exact $\mathbf{u} = 0$. The
 392 ligament stiffness and pretension terms are required for consistency with the assembled solver.
 393 For one spring segment of length L , direction $\mathbf{e}$, assigned stiffness k and pretension T , its
 394 directional stiffness is

$$395 \quad \mathbf{W} = k\mathbf{e}\mathbf{e}^T + (T/L)(\mathbf{I} - \mathbf{e}\mathbf{e}^T). \quad (8)$$

396 This matrix is assembled with opposite off-diagonal blocks at its two endpoints. It is a linearized
 397 spring representation without tension-only switching.

398 The surface measure obeys $dS_i = J_{surf,i} dS_0$, where

$$399 \quad J_{surf,i} = J_i \|\mathbf{F}_i^{-T} \mathbf{n}_0\|, \quad \mathbf{t} = \frac{\mathbf{F}^{tot}}{\int_{\Gamma_{load,0}} J_{surf,i} dS_0}. \quad (9)$$

400 Thus the applied physical force integral equals $\mathbf{F}^{tot}$. The penalty and load integrals use the
 401 corresponding mapped surface measures. The force vectors are listed in Table 2; their global
 402 axes are not assumed identical to the PCA reporting axes below.

403 A.3 Unloaded and loaded surfaces

404 Let $\mathbf{X}_j$ be surface template nodes. The two point clouds used for kinematics are

$$405 \quad \mathbf{p}_{ij}^0 = \mathbf{X}_j + \phi_i(\mathbf{X}_j), \quad \mathbf{p}_{i\ell j}^1 = \mathbf{p}_{ij}^0 + \mathbf{u}_{i\ell}(\mathbf{X}_j). \quad (10)$$

406 Both use identical topology and point correspondence. Here “unloaded” denotes the geometric
 407 reference with zero stored elastic displacement, not a separately equilibrated pretension-only

state. The loaded field includes both the prescribed external load and the ligament pretension right-hand side. Comparing $\mathbf{X}_j$ with $\mathbf{p}_{ilj}^1$ would incorrectly include subject morphology in the measured motion. Geometry was reconstructed with the original two-stage interpolation: template-to-FE mapping (10 neighbors, smoothing 50) followed by FE-to-surface mapping (10 neighbors, zero smoothing). Saved elastic fields use the latter interpolation. Cached sparse interpolation weights implement the same linear RBF evaluation; numerical comparisons against direct RBF evaluation check this equivalence.

A.4 Pelvis-aligned reporting frame

The origin $\mathbf{o}$ is the centroid of the unloaded surface. The initial ML axis is the dominant eigenvector of the point-cloud covariance, with deterministic sign. The symmetry-plane routine uses this PCA approximation; it does not optimize reflection symmetry. Project the centered points onto its orthogonal plane and take the leading in-plane singular vector. Orthogonalization gives $\hat{\mathbf{z}}$, followed by

$$\hat{\mathbf{y}} = \frac{\hat{\mathbf{z}} \times \hat{\mathbf{x}}}{\|\hat{\mathbf{z}} \times \hat{\mathbf{x}}\|}, \quad \hat{\mathbf{z}} = \hat{\mathbf{x}} \times \hat{\mathbf{y}}, \quad \mathbf{P} = [\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}}]. \quad (11)$$

The ML and AP axes are simultaneously reversed if needed so that ML points from the left to right ilium centroids. This construction is orthonormal and right-handed but does not independently anchor the AP/CC signs to anatomical landmarks. Absolute components are therefore used for comparisons. No clinical nutation sign is inferred from the stored α_x sign.

A.5 Affine correction, regions and rigid fits

Fit the sacral point correspondence by ordinary least squares,

$$(\mathbf{A}_S, \mathbf{b}_S) = \arg \min_{\mathbf{A}, \mathbf{b}} \sum_{j \in S} \|\mathbf{p}_j^1 - \mathbf{A} \mathbf{p}_j^0 - \mathbf{b}\|^2, \quad \mathbf{q}_j = \mathbf{A}_S^+(\mathbf{p}_j^1 - \mathbf{b}_S), \quad (12)$$

where $+$ denotes the Moore–Penrose pseudoinverse. This removes fitted affine sacral deformation, including stretch and shear; it is not only a change of rigid observer frame.

For each side, first restrict both point clouds to that side of the reporting-frame midplane (falling back to the entire bone if this set is empty). Select sacral and iliac points within 5 mm of the opposing restricted point cloud. If fewer than 80 points qualify, use up to the nearest 80 available points. Distances are between discrete points rather than continuous articular surfaces.

435 For each bone ROI, the rigid fit maps $\mathbf{p}^0$ to $\mathbf{q}$. With centered row matrices $\tilde{\mathbf{P}}_B, \tilde{\mathbf{Q}}_B$,

$$436 \quad \tilde{\mathbf{P}}_B^T \tilde{\mathbf{Q}}_B = \mathbf{U} \Sigma \mathbf{V}^T, \quad \mathbf{R}_B = \mathbf{V} \text{diag}(1, 1, \det(\mathbf{V} \mathbf{U}^T)) \mathbf{U}^T, \quad \mathbf{t}_B = \bar{\mathbf{q}}_B - \mathbf{R}_B \bar{\mathbf{p}}_B^0. \quad (13)$$

437 The fit is repeated after excluding the largest 10% residuals, using two trimming iterations.

438 Trimming reduces outlier influence but does not validate the ROI as a physical contact patch.

439 A.6 Relative rotation and translation

440 For sacral and iliac fits S, I ,

$$441 \quad \mathbf{R}_{rel} = \mathbf{R}_S^T \mathbf{R}_I, \quad \mathbf{R}_{loc} = \mathbf{P}^T \mathbf{R}_{rel} \mathbf{P} = \mathbf{R}_z(\gamma_z) \mathbf{R}_y(\beta_y) \mathbf{R}_x(\alpha_x). \quad (14)$$

442 The fixed-axis xyz extraction, for zero-based matrix indices away from gimbal lock, is

$$443 \quad \alpha_x = \text{atan2}(R_{21}, R_{22}), \quad \beta_y = \arcsin[-\text{clip}(R_{20}, -1, 1)], \quad \gamma_z = \text{atan2}(R_{10}, R_{00}). \quad (15)$$

444 At $\beta_y = \pm\pi/2$, choose $\gamma_z = 0$ and $\alpha_x = \text{atan2}[-\text{sign}(R_{20})R_{01}, R_{11}]$. In implementation, the
 445 equivalent stable expression $\beta_y = \text{atan2}(-R_{20}, \sqrt{R_{00}^2 + R_{10}^2})$ is used, with the singular branch
 446 restricted to vanishing cosine. The singular branch is included for completeness; physiological
 447 small-angle outputs should remain far from it. Matrix reconstruction tests cover both branches.

448 For the unloaded sacral ROI centroid $\mathbf{p}_c$, the reported displacement is

$$449 \quad \mathbf{d} = \mathbf{P}^T [(\mathbf{R}_I - \mathbf{R}_S) \mathbf{p}_c + \mathbf{t}_I - \mathbf{t}_S]. \quad (16)$$

450 It evaluates the difference of two fitted maps at a common reference point, expressed in the
 451 unloaded pelvic axes after affine correction. It is not the translational parameter of $T_S^{-1}T_I$. In
 452 fact, with $\mathbf{t}_{rel} = \mathbf{R}_S^T(\mathbf{t}_I - \mathbf{t}_S)$,

$$453 \quad \mathbf{d} = \mathbf{P}^T \mathbf{R}_S [(\mathbf{R}_{rel} - \mathbf{I}) \mathbf{p}_c + \mathbf{t}_{rel}]. \quad (17)$$

454 This identity makes explicit the distinction between the relative rigid transform and the reported
 455 point-displacement difference. It is neither the distance between opposing articular surfaces nor

a displacement along a surface normal. The bilateral summaries are

$$|\boldsymbol{\theta}| = \frac{1}{2} \sum_{s=L,R} (\alpha_{x,s}^2 + \beta_{y,s}^2 + \gamma_{z,s}^2)^{1/2}, \quad |\mathbf{d}| = \frac{1}{2} \sum_{s=L,R} \|\mathbf{d}_s\|, \quad (18)$$

$$|d_k| = \frac{1}{2} (|d_{k,L}| + |d_{k,R}|), \quad A_d = \left| \|\mathbf{d}_L\| - \|\mathbf{d}_R\| \right|. \quad (19)$$

Angles are converted to degrees before reporting. The exact principal rotation angle would instead be $\arccos[(\text{tr } \mathbf{R}_{rel} - 1)/2]$ with numerical clipping. The Cardan norm approximates it at small angles and depends on the chosen axes and sequence.

A.7 Size and conditional scaling

The archived allometry workflow sums volumes of the selected left ilium, right ilium and sacrum surface bodies, using mapped surface geometry and converting mm^3 to cm^3 . These are whole-bone geometric volumes, not birth-canal volume or a direct integral of cortical tissue. The size factor is $s_i = (V_i/V_{template})^{1/3}$. Therefore a log-volume slope b becomes $3b$ under log scale, with identical fitted responses and tests.

For the sex-interaction log model, $b_M = \beta_v$ and $b_F = \beta_v + \beta_{v:s}$. The female slope variance is

$$\text{Var}(b_F) = \text{Var}(\beta_v) + \text{Var}(\beta_{v:s}) + 2 \text{Cov}(\beta_v, \beta_{v:s}). \quad (20)$$

Reported intervals use this full covariance. Exponentiating a fitted log response gives a conditional geometric mean, not an unbiased arithmetic mean. Figure 7 reports slopes from this model. Supplementary Figure S1 shows its conditional geometric-mean curves with age and the morphometric covariates set to their cohort means.

A.8 Statistical estimands and uncertainty

All logarithms are natural logarithms of numerical values in the stated units: $\log V$ abbreviates $\log[V/(1 \text{ cm}^3)]$, and $\log Y$ uses 1 mm or 1 degree as the reference unit. Changing these reference units changes intercepts, but not the sex-specific volume exponents. Standardized predictors are $z_i = (x_i - \bar{x})/s_x$, where s_x is the cohort sample standard deviation; for log volume, the logarithm is taken before standardization.

For design matrix $\mathbf{Z}$, OLS residuals e_i and leverage h_{ii} , the HC3 covariance estimator is

$$\widehat{\text{Cov}}_{HC3}(\hat{\boldsymbol{\beta}}) = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \text{diag} \left[\frac{e_i^2}{(1 - h_{ii})^2} \right] \mathbf{Z} (\mathbf{Z}^T \mathbf{Z})^{-1}. \quad (21)$$

For any coefficient contrast $\mathbf{c}$, its estimate is $\mathbf{c}^T \hat{\boldsymbol{\beta}}$ and standard error is $[\mathbf{c}^T \widehat{\text{Cov}}_{HC3} \mathbf{c}]^{1/2}$. Reported Wald intervals use the normal critical value (approximately 1.96). The allometry exponents are conditional elasticities: holding the other predictors fixed, multiplying volume by $a > 0$ multiplies the fitted geometric-mean response by a^b .

For a paired contrast Δ_i , the reported effect is median _{i} Δ_i , not the difference between two marginal medians. With n_* nonzero differences and $B = \sum_i \mathbb{I}(\Delta_i > 0)$, the two-sided exact sign test uses $B \sim \text{Binomial}(n_*, 1/2)$ under the null of balanced signs. For a continuous difference distribution this is a test of zero median; excluding exact zeros conditions on nonzero differences. Bootstrap intervals are percentile intervals obtained by resampling subjects, preserving the paired observations. They are pointwise intervals, whereas the reported q values apply Benjamini–Hochberg adjustment within the specified test families; these two summaries need not cross their respective thresholds together.

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Author contributions

CRedit authorship contribution statement. Petr Henyš: Conceptualization; Methodology; Software; Formal analysis; Investigation; Data curation; Visualization; Funding acquisition; Writing – original draft; Writing – review & editing. Niels Hammer: Supervision; Writing – review & editing. Final author approval of this revised version is pending.

Conflict of interest statement

The authors declare no conflicts of interest.

Supporting information

Supporting Information provides statistical definitions, component contrasts, variance ratios, sex models, the scale reparameterization, and extraction sensitivity diagnostics.

Data Availability Statement

The manuscript working package contains corrected subject-level endpoints, statistical tables and figures. The original FE displacement archives are retained separately. A persistent public or reviewer-accessible repository link must be supplied before submission.

Code Availability Statement

The working package includes scripts for corrected endpoint extraction, statistical analysis and figure generation. A versioned code archive and environment specification must accompany the data repository before submission.

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